seem to be worth studying, so that biologists interested in this kind of work may find it worth while to watch the biotype in its various stages of development.

[1] Pringsheim, E. G., and Robinow, C. F., *J. Gen. Microbiol.*, **1**, 267 (1947).
[2] Pringsheim, E. G., *J. Gen. Microbiol.*, **5**, 124 (1951).
[3] Baker, F., *Ann. App. Biol.*, **30**, 230 (1943).
[4] Pringsheim, E. G., *Arch. Microbiol.*, **16**, 18 (1951).
[5] Küster, E., *Arch. Protistenk.*, **2**, 351 (1908).
[6] Pringsheim, E G., *J. Protozool.*, **2**, 137 (1945).
[7] Harold, R., and Stanier, R. Y., *Bact. Rev.*, **19**, 49 (1955).

THE QUESTION OF CORRELATION BETWEEN PHOTONS IN COHERENT LIGHT RAYS

By Prof. ERIC BRANNEN and H. I. S. FERGUSON
Department of Physics, University of Western Ontario, London, Canada

CONSIDERABLE interest has been aroused in this question, especially since the experiments of R. Hanbury Brown and R. Q. Twiss[1] showed a correlation, unexpected by many, between the current outputs from two photoelectric detectors viewing coherent light rays. On the other hand, A. Ádám, L. Jánossy and P. Varga[2] performed a similar experiment using "an amplifier with a resolving time of 2 μsec." and observed no such correlation.

In this laboratory we have been performing coincidence experiments[3] on photons emitted in cascade transitions from optical sources using apparatus of resolving time $2\tau = 10^{-8}$ sec. With this apparatus[4], which is now standard in nuclear physics, time delays can be inserted in either photomultiplier channel, and so a distinct answer to the question of time correlation can be obtained. Our apparatus has observed true coincidences in the above case of optical transitions in cascade. This is of prime importance, since our negative results with coherent light rays can have meaning only if the apparatus has been shown to be capable of, and efficient in, observing 'true' coincidences.

The experiment reported here was designed to duplicate so far as possible the optical arrangement of Brown and Twiss, up to and including the photomultipliers. It is in the electronic detection of coincidences between the outputs of the photomultipliers in which we differ from them. Furthermore, our apparatus compares pulses due to the detection of individual photons by the photomultipliers, rather than an integrated output due to large numbers of photons. We also feel that the Brown and Twiss experiment does not sufficiently reject the possibility that the correlation they observed in the 'photocathodes superimposed' position was due to rapid intensity fluctuations in the source. It seems probable that classical wave theory would predict correctly this case, since both photomultipliers would be observing the same part of the source, and since the fluctuating part of the photocurrent was observed. Furthermore, no correlation would be observed in the 'photocathodes not superimposed' position if their movable photomultiplier then viewed another region of the source, which was comparatively steady, or the intensity fluctuations of which were not correlated with the previous region. Although we do not feel competent to answer the question, it occurs to us that the radio telescope of Brown and Twiss could be working in a similar manner if the galactic source of radio noise was fluctuating in intensity with different regions fluctuating, for example, independently. It should here be emphasized that random coincidences are evaluated in our experiment by insertion of a time delay in one channel much greater than the resolving time of the apparatus. This means that such macroscopic intensity fluctuations in the source would not affect the results of our experiment.

The experimental arrangement is shown in Fig. 1. S is a high-pressure Hilger mercury arc. The image of the arc is focused by means of an $F2$ Zeiss Biotar lens L on to a small pinhole H of diameter 0·25 mm. The light is filtered by a Baird interference filter (λ4358 A.) and the appropriate Wratten filter, at F, so that essentially monochromatic light falls on the pinhole. It was found that a good lens free of aberrations is essential in order to get a uniform beam free of striations falling on the half-silvered mirror M. The light transmitted by the mirror is detected by a $1P21$ photomultiplier P_1, and that reflected is detected by a similar photomultiplier P_2. The areas of the two photocathodes are limited by identical rectangular apertures A_1 and A_2, 4 mm. by 8 mm. Single photons detected by the photomultipliers give rise to voltage pulses which are shaped (5×10^{-9} sec. duration), and delayed by coaxial cables L_1 and L_2. They are mixed in the coincidence stage and an output pulse obtained if they are in time coincidence at the coincidence stage. In the diagrams, positive delay refers to P_1 pulses delayed with respect to P_2 pulses, and negative delay vice versa. The total coincidence rate C as a function of delay, and the counting rates C_1 and C_2 in each channel, were recorded. Throughout the experiments C_1 and C_2 remained sensibly constant.

The following set of experiments was carried out in order to test the equipment and to test for correlation between photons: (*a*) cable attenuation corrections; (*b*) coincidence rate as a function of

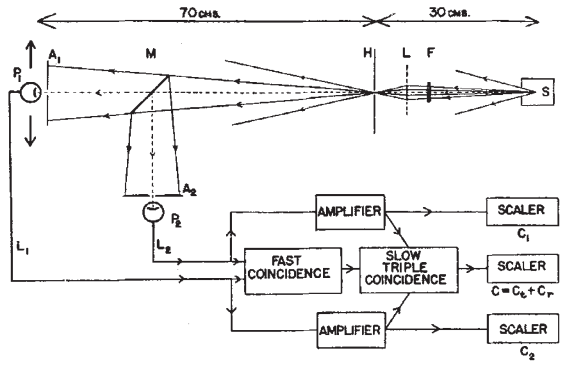


Fig. 1. Simplified diagram of apparatus

©1956 Nature Publishing Group

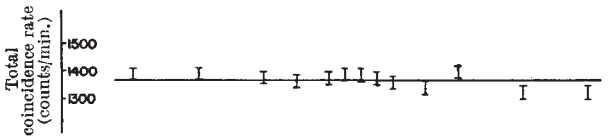


Fig. 2. Total coincidence rate versus delay, 'photocathodes superimposed'

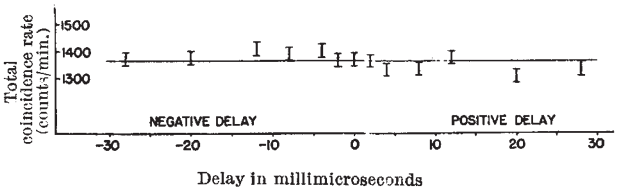


Fig. 3. Total coincidence rate versus delay, 'photocathodes not superimposed'

delay, 'photocathodes superimposed'; (*c*) coincidence rate as a function of delay, 'photocathodes not superimposed'; (*d*) coincidence rate as one of the photomultipliers scans the beam; (*e*) coincidence rate as intensity of the source varied; (*f*) prompt resolution curve for the coincidence apparatus. In all the experiments the apparatus was functioning stably and consistently.

The cable attenuation corrections were obtained with the photomultipliers observing independent sources, and in no case are they greater than several per cent. These have been applied to all the following results.

With P_1 and P_2 in the 'photocathodes superimposed' position, the total coincidence rate as a function of delay was recorded and the results of Fig. 2 obtained. It is noted that within the statistical error of a probable error of 1 per cent, there was no increase in coincidence counting rate at zero delay relative to long delays.

The 'photocathodes not superimposed' position was obtained by moving P_2 laterally a distance of 12 mm., which is equivalent to three times the slit width. The total coincidence rate as a function of delay was again recorded, and the results plotted in Fig. 3. Again there was no increase in coincidence counting rate at zero delay relative to long delays. It is readily apparent that the two sets of results presented in Figs. 2 and 3 are indistinguishable.

In order to ensure that the image position had not been missed, P_1 was moved in steps small compared to the slit width and the total coincidence rate recorded for zero delay. The flat curve of Fig. 4 was obtained. The beam was quite uniform as shown by the total side-channel count from P_1, which remained constant (variation less than 1 per cent) over the whole width of the beam. There is no increase in coincidence counting rate at the 'photocathodes superimposed' position.

A further check was carried out—the intensity of the light source was varied, and the coincidence rate at zero delay, and the side-channel rates, recorded.

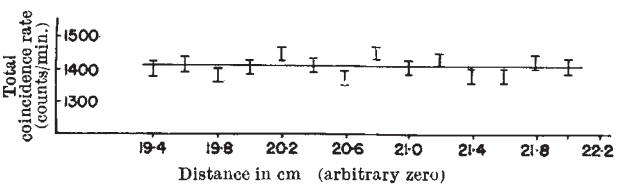


Fig. 4. Total coincidence rate at zero delay versus position. (Photomultiplier P_2 moving in steps across the image)

The coincidence rate C was proportional to the product C_1C_2 consistent with the coincidence rate being pure random[5] (random rate $C_r = 2\tau C_1C_2$).

Finally, to ensure that the apparatus was functioning properly, resolution curves (Fig. 5) were obtained before and after the previous experiments. Stilbene crystals were placed in front of the photomultipliers and a cobalt-60 source (1·17 and 1·33 MeV. gamma rays in cascade) used as a source of prompt radiation.

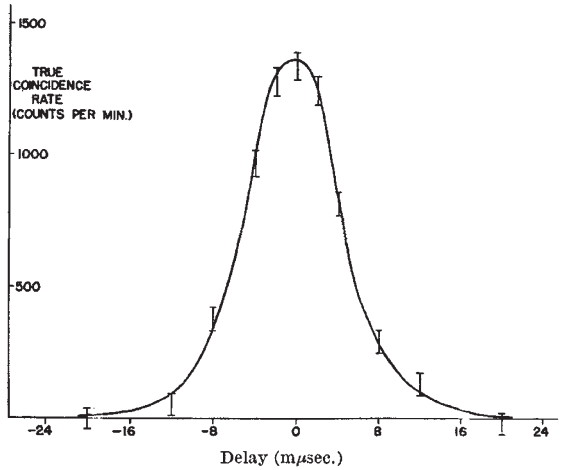


Fig. 5. Prompt resolution curve

From these experiments one can place an upper limit on the fraction of photons acting in an anomalous manner. The side-channel rates C_1 and C_2 were each $(5{\cdot}00 \pm 0{\cdot}05) \times 10^4$ counts per second. The coincidence rate was $24{\cdot}0 \pm 0{\cdot}4$ counts per second. Even if one goes to eight times the probable error there can be no more than three true coincidences per second. Since the 1*P*21 photomultipliers have a photoelectric efficiency of about 10 per cent at a wave-length of 4358 A., this means that the probability is less than 1 in 10^7 that more than 0·03 per cent of the photons detected are in true coincidence. (Ádám *et al.*[2] estimated that less than 0·6 per cent of the photons are in time coincidence allowing a deviation of three times the probable error. In our case, using three times the probable error, less than 0·01 per cent are in time coincidence.)

In conclusion, our experiments indicate that there is no correlation (less than 0·01 per cent) between photons in coherent light rays.

We are indebted to Dr. W. Wehlau, in the Astronomy Department at this University, who brought this controversial matter to our attention. Discussion with him on this and allied subjects has proved most valuable.

Note added in proof. It would appear to the authors, and also to Prof. Jánossy (private communication), that if such a correlation did exist, it would call for a major revision of some fundamental concepts in quantum mechanics. This was, of course, the reason why these experiments were performed.

[1] Hanbury Brown, R., and Twiss, R. Q., *Nature*, **177**, 27 (1956).
[2] Ádám, A., Jánossy, L., and Varga, P., *Acta Phys. Hungary*, **4**, No. 4, 301 (1955).
[3] Brannen, E., Hunt, F. R., Adlington, R. H., and Nicholls, R. W., *Nature*, **175**, 810 (1955).
[4] Bell, R. E., Graham, R. L., and Petch, R. E., *Canad. J. Phys.*, **30**, 35 (1952).
[5] Dunworth, J. V., *Rev. Sci. Instrum.*, **11**, 167 (1940).

©1956 Nature Publishing Group